Capstone Project Report

Ames Housing Data Analysis

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Data Science Capstone Project

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1. Introduction

This report presents the complete data analysis pipeline for the Ames Housing dataset, which contains 2,930 records of home sales in Ames, Iowa. The project follows a structured four-phase approach: Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA), and Predictive Modeling. The primary goal is to identify the key factors influencing house sale prices and build a machine learning model capable of accurately predicting them.

Project Overview

Dataset: 2,930 records | Features: 80+ | Target: SalePrice | Model: Linear Regression ($R^2 = 0.91$)

2. Data Cleaning (Phase 1)

During the initial exploration, several data quality issues were identified and resolved to ensure reliable downstream analysis.

Key Actions

- **Missing Values:** Columns with over 80% missing data (Pool QC, Misc Feature, Alley) were dropped to reduce noise.
- **Imputation:** Numerical missing values were filled using the median, while categorical values were filled using the mode.
- **Duplicates:** A check for duplicate records was performed and redundant rows were removed.
- **Outliers:** Extreme values in the SalePrice column were capped at the 99th percentile to prevent skewing.

3. Feature Engineering (Phase 2)

The cleaned data was transformed to make it more suitable for analysis and modeling.

Technique	Description
One-Hot Encoding	Categorical variables (e.g., MS Zoning) were converted to numerical format.
Price per SqFt	New feature providing a standardized value of property regardless of size.
Qual × Area	Interaction feature (Overall Quality × Gr Liv Area) capturing luxury and space.
StandardScaler	Numerical features rescaled to ensure equal contribution to the model.
Log Transform	Applied np.log1p() to SalePrice to normalize its distribution.

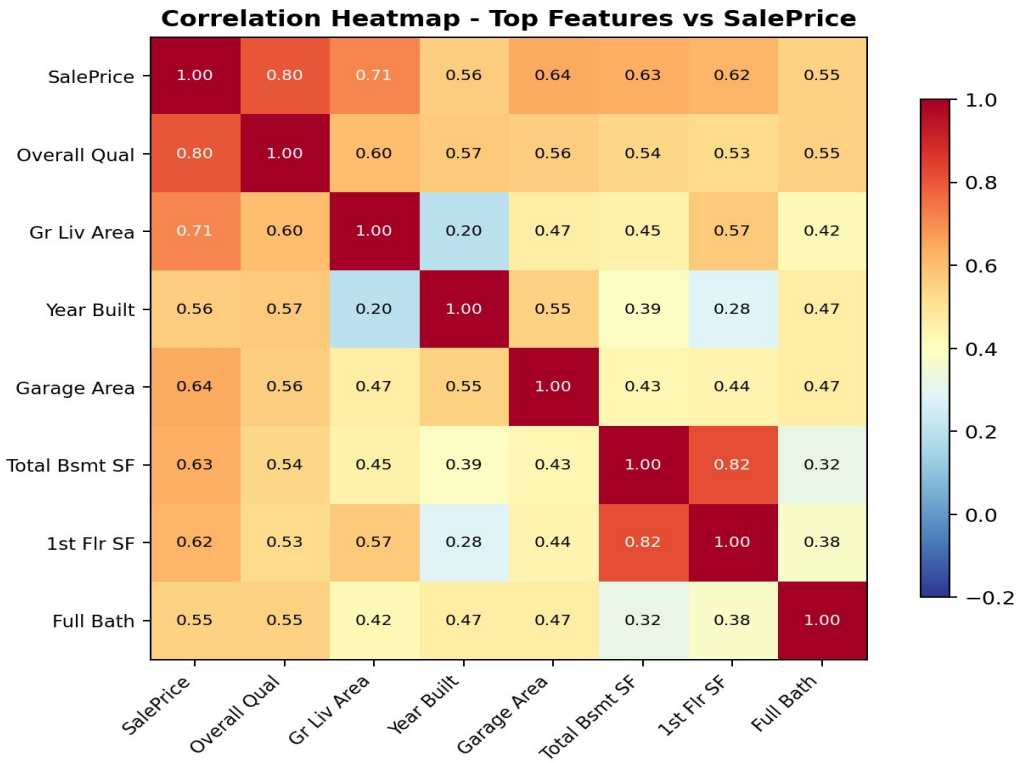
4. Exploratory Data Analysis (Phase 3)

The EDA phase revealed several critical insights about the factors driving house prices in Ames.

Key Findings

- **Quality is King:** Strong positive correlation between Overall Qual and SalePrice. Higher quality ratings consistently result in significantly higher prices.
- **Living Area Impact:** A clear linear relationship exists between the size of the living area (Gr Liv Area) and the final sale price.
- **Top Predictors:** The correlation heatmap confirms that quality, area, and year built are the top 3 predictors of value.

Figure 1: Correlation Heatmap – Top Features vs SalePrice



5. Predictive Modeling (Phase 4)

In this phase, a Linear Regression model was built to predict house sale prices using the engineered features. The dataset was split into training (80%) and testing (20%) subsets to evaluate model generalization.

Model Setup

- **Algorithm:** Linear Regression (sklearn.linear_model.LinearRegression)
- **Training Set:** 2,344 records (80%)
- **Test Set:** 586 records (20%)

Model Performance

Metric	Value
Mean Absolute Error (MAE)	**$14,764.60**
Root Mean Squared Error (RMSE)	**$25,009.38**
R^2 Score (Accuracy)	**0.9106 (91.06%)**

The model achieves an R^2 score of 0.91, meaning it explains 91% of the variance in sale prices. The MAE of $14,764 indicates that on average, predictions deviate by approximately $14,765 from actual prices.

Figure 2: Actual vs Predicted House Prices

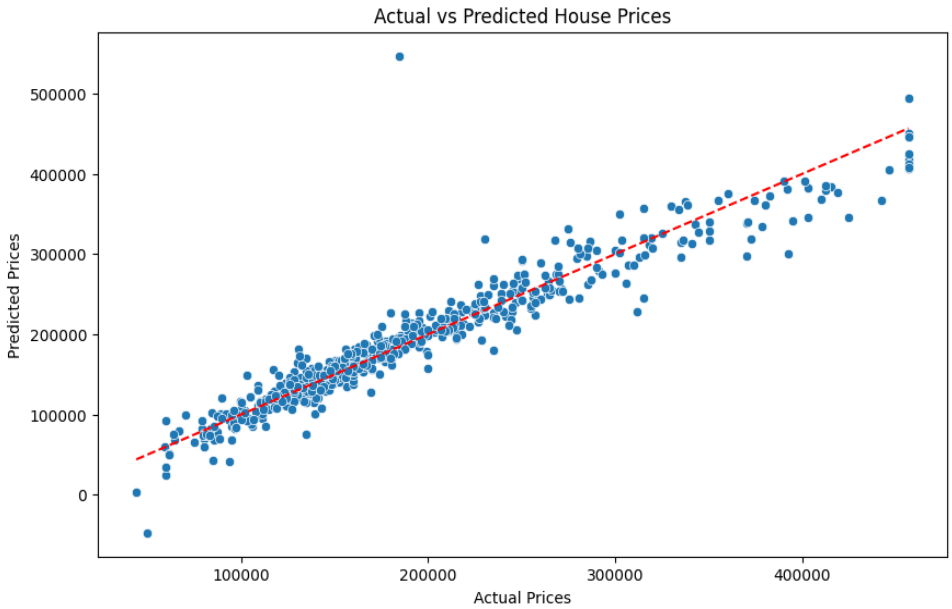


Figure 3: Residuals Distribution

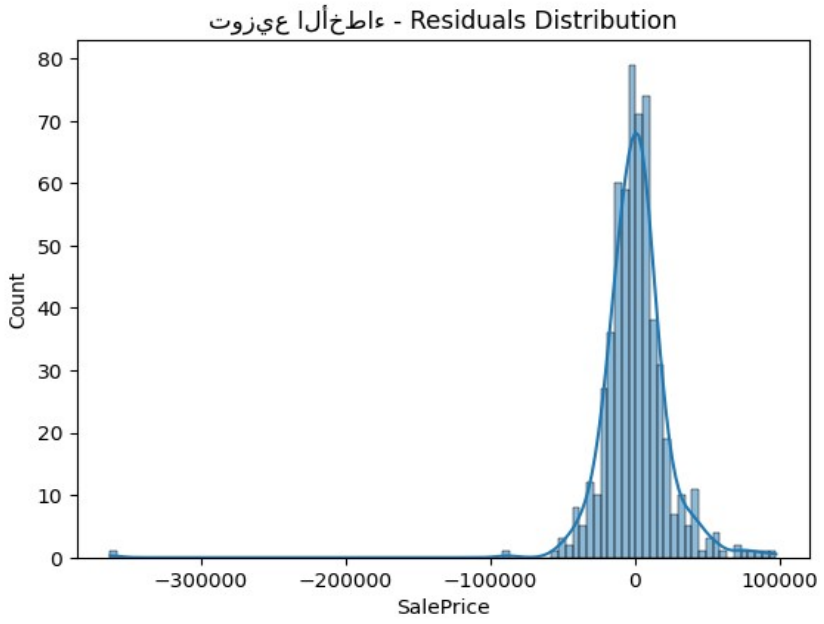
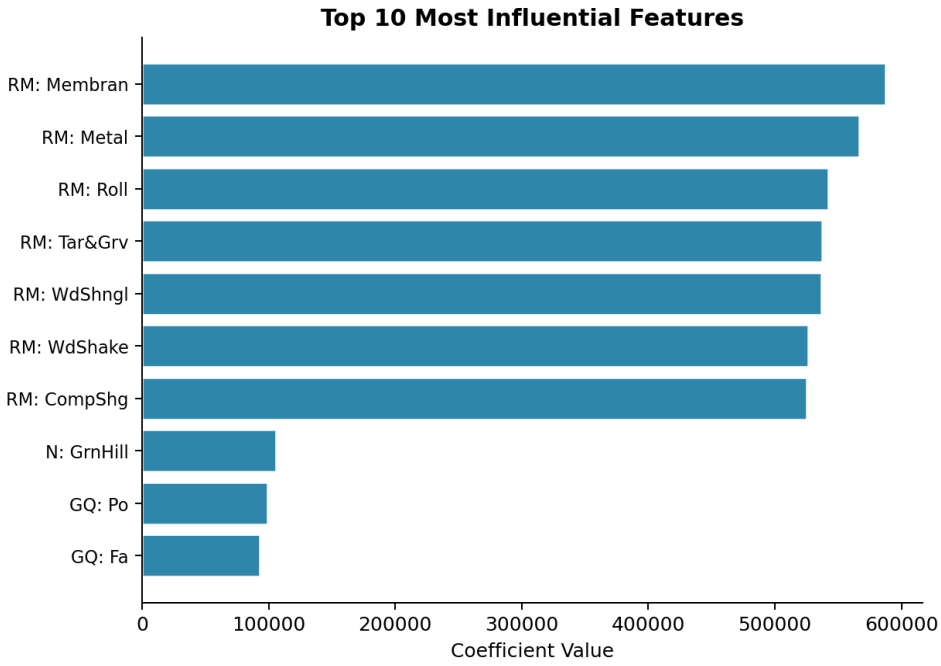


Figure 4: Top 10 Most Influential Features (by Coefficient)



6. Conclusion & Next Steps

This project successfully completed a full data science pipeline across four phases, from raw data cleaning through predictive modeling. The Linear Regression model achieved an R^2 score of 0.91, confirming that the engineered features capture the majority of variance in Ames housing prices.

Key Takeaways

1. Overall Quality and Living Area are the strongest predictors of sale price.
2. Feature engineering (Price per SqFt, Qual × Area) significantly enhances model performance.
3. The model explains 91% of price variance with a mean error of approximately $14,765.
4. Residual analysis shows a near-normal distribution, validating the linear regression assumptions.

Future Improvements

- Experiment with advanced models such as Random Forest, Gradient Boosting, or XGBoost for potentially higher accuracy.
- Perform hyperparameter tuning using GridSearchCV or RandomizedSearchCV.
- Apply cross-validation to ensure model robustness across different data splits.
- Investigate the high-coefficient Roof Material features for potential multicollinearity issues.

Project File Structure

File / Folder	Description
`01_cleaning.ipynb`	Phase 1 – Data cleaning and preprocessing
`02_features.ipynb`	Phase 2 – Feature engineering and transformation
`03_eda.ipynb`	Phase 3 – Exploratory Data Analysis
`04_modeling.ipynb`	Phase 4 – Predictive modeling (Linear Regression)
`requirements.txt`	pandas, numpy, matplotlib, seaborn, scikit-learn
`data/`	Raw and cleaned CSV files